Formal Requirements Of Structure

1. Be in NeurIPS style: https://neurips.cc/Conferences/2015/PaperInformation/StyleFiles
2. Core paper length: 5-10 pages. Unlimited pages for references and technical appendices.
    a. The core paper must tell the main story including situating your project within the relevant literature as well as having key figures.
    b. Must have a short abstract that concisely summarises your project findings.
    c. Technical appendices should be used to give supplemental details that are ancillary to the main story you are telling and just won't fit within the 10 page limit. *What is supplemental and what is ancillary will depend on what the main story is for your project*. e.g. for an optimization-centric project, details of hyperparameter choice for the optimizer, etc. can be central. These are likely to be ancillary for a project on Toy models for ICL.
    d. We expect people will extend and update their proposal to form this report.
3. The final report will have to be accompanied by a released github for your project that permits replicability of your work.
    a. At the minimum, there needs to be a way for a reviewer to replicate all your plots and tables easily.
    b. If you are using a publicly accessible dataset, you don't need to include your datasets. Just have code that can fetch them.
    c. For certain Projects (rare cases of ones from other areas — never for optimization, toy ICL, or interpretability) where the datasets are too vast or sensitive to be released, you need an explanation of this. For them, you should have at least a tiny dataset that can be used to verify that the training and testing code works.

Content Requirements

1. Everything that we required from the Project Proposal, in expanded and more detailed form.
    a. Your literature review should not just crisply position your work in the context of the literature, it should also include a detailed discussion of related work exploring ideas similar to what you are exploring. We expect most projects to therefore have at least 2-3 full pages discussing the relevant literature.
    b. Expanded methods section that makes it clear what exactly you are doing. In the draft report, mark things that are still in progress.
2. Clear self-contained mathematics and fully labeled diagrams that result in no ambiguity in what you are doing. It should be possible to read without having to reference other papers to understand what you are doing. Use code-snippets or pseudocode when those are the clearest way to explain something.

3. Tables with clear settings for hyperparameter values along with how/why those values were chosen. It is fine if some were chosen based on selections in other papers, but you need to cite and explain why you chose that value. For others, there should be fully described experiments backing up that choice — these can be in technical appendices if this is ancillary to your main story.
4. Well labeled plots with error bars as appropriate. The scale (linear, log, log-log, offsets, etc.) should be chosen consciously and explained. Essentially every project will have a training curve of some sort. Essentially every project will have some sort of validation loss. Error bars should be clearly explained — how were they generated and why are they the right tool for what you want to understand.
5. Your experiments need to be controlled to the extent practicable — if you're comparing to something in the literature, you should be changing one thing at a time if you can. Commentary on this aspect is required.
6. If there are techniques that you are using that weren't covered in the course, they should be explained at a level so your peers can understand them.

Draft vs Final

The point of the draft is to get something that peer reviewers can read and give you feedback on. It is intended to help you make your final project that much better — because we have high expectations from you. If an experiment is incomplete or in progress, report on what you have so far. Point out uncertainties clearly.

However, we expect drafts to be well written. We encourage you to use LLMs to get substantive critiques of your reports before submitting them. Incorporate their suggestions. Posting prompts that help get good feedback in this thread is a valid form of class participation.